Network Performance Monitoring and Traffic Analysis System

Course Code: CSA0927
Course Name: Programming in Java

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Introduction

- Network performance monitoring helps capture and analyze network traffic.
- The system monitors packets transmitted between devices.
- The project is developed using Java with a user-friendly graphical interface.
- It captures packet information such as source IP, destination IP, protocol, packet size, and timestamp.
- The system helps users understand network communication and traffic flow.

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Objective

•To develop a Java-based system that captures, analyzes, and monitors network traffic in real time for efficient network performance monitoring.

- To develop a Java-based network packet monitoring system.
- To capture network packets in real time.
- To analyze important packet information.
- To identify protocols such as TCP, UDP, ICMP, and ARP.

LITERATURE SURVEY

S.No	**Author(s)**	**Year**	**Title**	**Limitation**
1	Sharma	2023	Real-Time Network Traffic Analysis Using Packet Sniffing	Difficulty handling large volumes of network traffic.
2	Kumar & Singh	2023	Network Security Monitoring Through Packet Inspection	High CPU and memory consumption during continuous monitoring.
3	Patel	2024	Java-Based Network Packet Analyzer	Limited support for advanced protocol analysis.
4	Xing & Lee	2024	Packet Capture and Traffic Monitoring System	Lacks an intuitive graphical user interface.
5	Lee	2025	Intelligent Network Traffic Analysis Framework	Reduced performance in high-speed network environments.

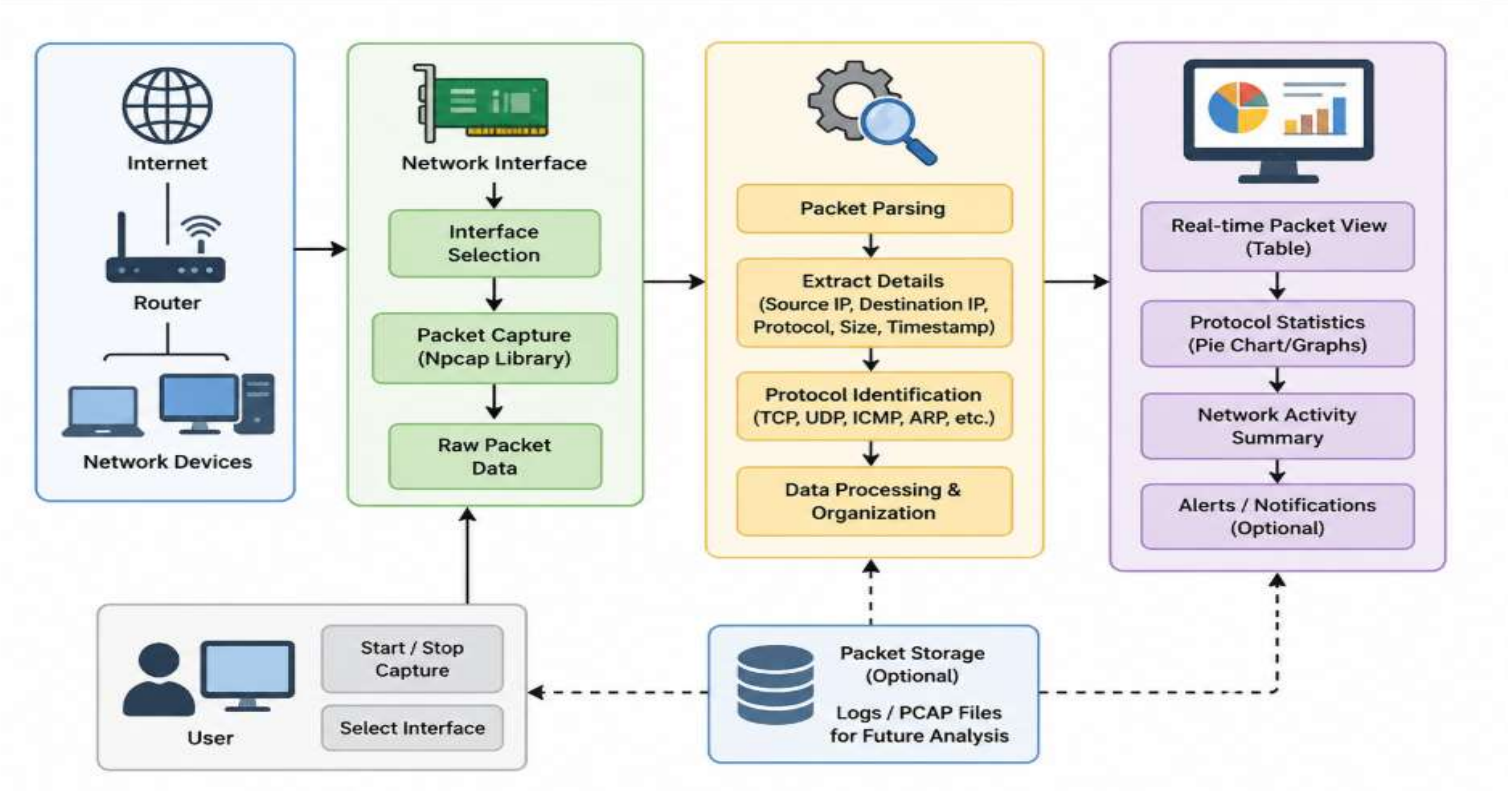
Problem Statement

- Modern networks generate a large amount of traffic that needs to be monitored efficiently.
- Existing network monitoring tools can be complex for beginners.
- Some monitoring tools require high system resources.
- Understanding packet-level communication can be difficult without a simple interface.
- Therefore, there is a need for a simple Java-based system for real-time packet capture and traffic analysis.

Proposed System

- The proposed system captures network packets from a selected network interface.
- Captured packets are processed and analyzed in real time.
- Important information such as source IP, destination IP, protocol, packet size, and timestamp is extracted.
- Packets are classified based on protocols such as TCP, UDP, ICMP, and ARP.

System Architecture



Module 1: Packet Capture

- This module detects all available network interfaces in the system.
- The user can select a specific network interface to monitor.
- It captures incoming and outgoing network packets in real time.
- The captured packets contain raw network information that is collected for processing.
- The captured data is passed to the Packet Analysis module for further analysis.

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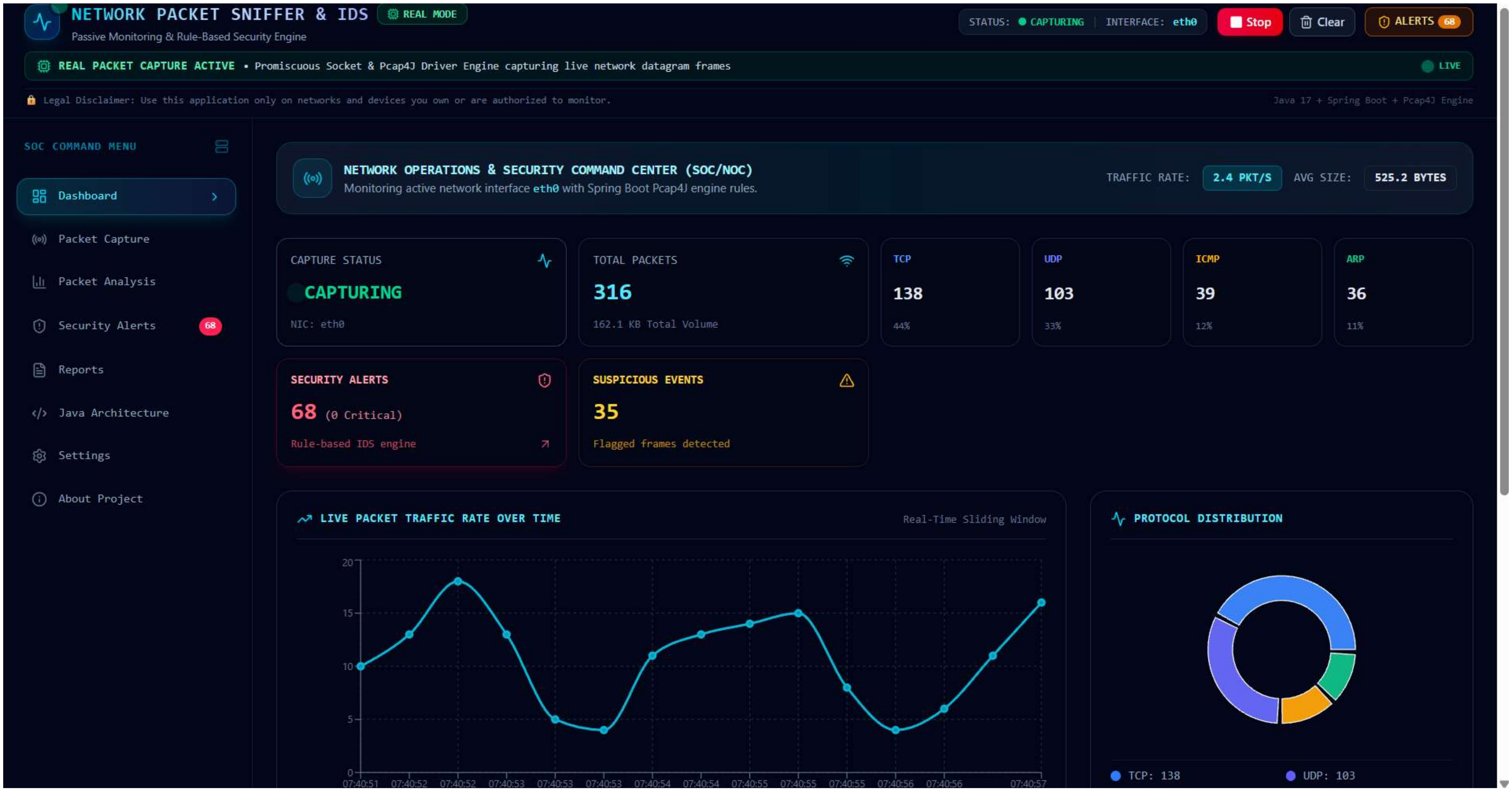
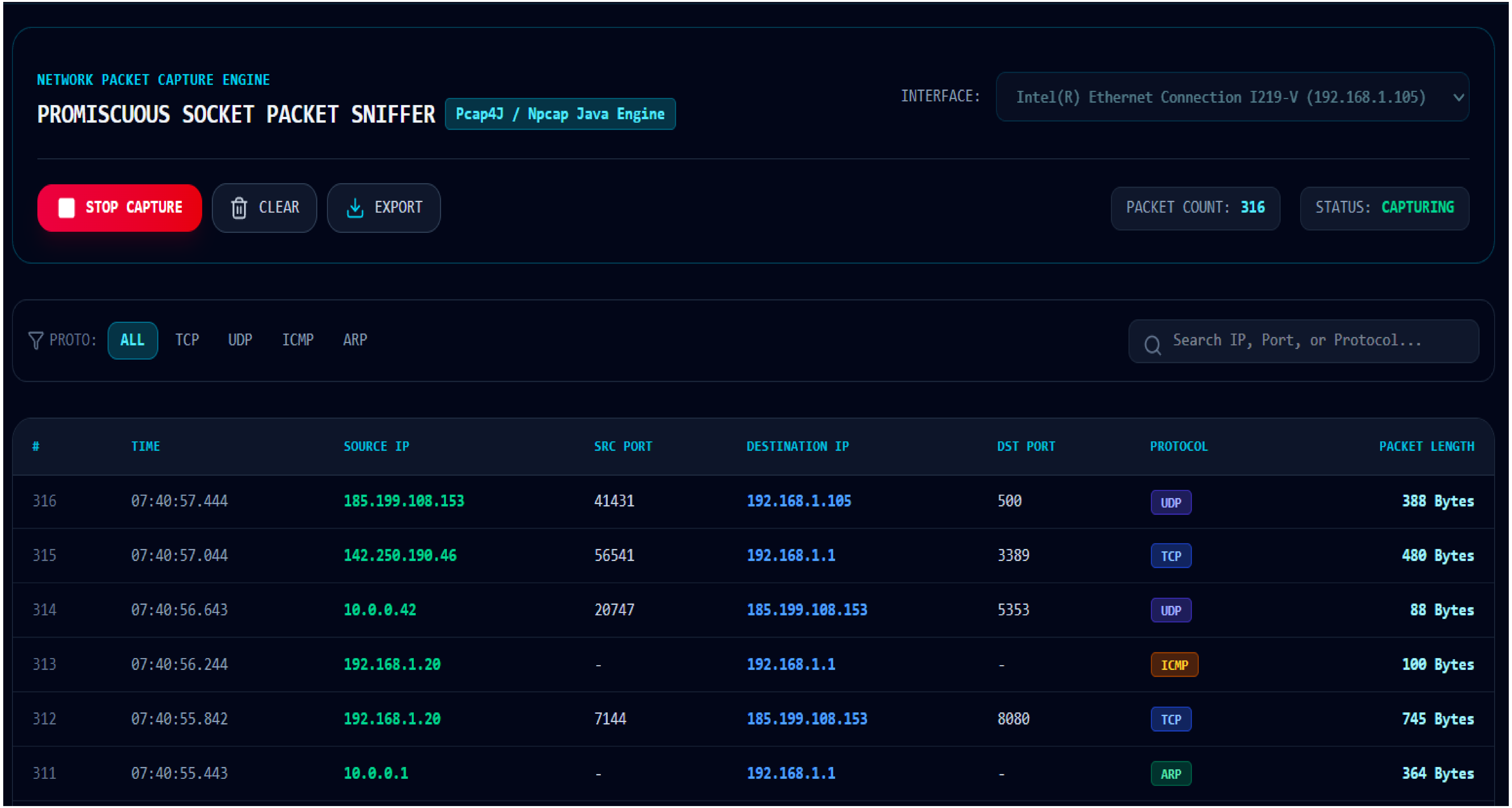

NETWORK PACKET SNIFFER & IDS
REAL MODE
Passive Monitoring & Rule-Based Security Engine
STATUS: CAPTURING | INTERFACE: eth0
Stop
Clear
ALERTS 68
REAL PACKET CAPTURE ACTIVE • Promiscuous Socket & Pcap4J Driver Engine capturing live network datagram frames
LIVE
Legal Disclaimer: Use this application only on networks and devices you own or are authorized to monitor.
Java 17 + Spring Boot + Pcap4J Engine
SOC COMMAND MENU
Dashboard
Packet Capture
Packet Analysis
Security Alerts 68
Reports
Java Architecture
Settings
About Project
NETWORK OPERATIONS & SECURITY COMMAND CENTER (SOC/NOC)
Monitoring active network interface eth0 with Spring Boot Pcap4J engine rules.
TRAFFIC RATE: 2.4 PKT/S
AVG SIZE: 525.2 BYTES
CAPTURE STATUS
CAPTURING
NIC: eth0
TOTAL PACKETS
316
162.1 KB Total Volume
TCP 138 44%
UDP 103 33%
ICMP 39 12%
ARP 36 11%
SECURITY ALERTS
68 (0 Critical)
Rule-based IDS engine
SUSPICIOUS EVENTS
35
Flagged frames detected
LIVE PACKET TRAFFIC RATE OVER TIME
Real-Time Sliding Window
20
15
10
5
0
07:40:51 07:40:52 07:40:52 07:40:53 07:40:53 07:40:53 07:40:54 07:40:54 07:40:55 07:40:55 07:40:55 07:40:56 07:40:56 07:40:57
PROTOCOL DISTRIBUTION
TCP: 138
UDP: 103


NETWORK PACKET CAPTURE ENGINE
PROMISCUOUS SOCKET PACKET SNIFFER
Pcap4J / Npcap Java Engine
INTERFACE:
Intel(R) Ethernet Connection I219-V (192.168.1.105)
STOP CAPTURE
CLEAR
EXPORT
PACKET COUNT: 316
STATUS: CAPTURING
PROTO:
ALL
TCP
UDP
ICMP
ARP
Search IP, Port, or Protocol...
TIME SOURCE IP SRC PORT DESTINATION IP DST PORT PROTOCOL PACKET LENGTH
316 07:40:57.444 185.199.108.153 41431 192.168.1.105 500 UDP 388 Bytes
315 07:40:57.044 142.250.190.46 56541 192.168.1.1 3389 TCP 480 Bytes
314 07:40:56.643 10.0.0.42 20747 185.199.108.153 5353 UDP 88 Bytes
313 07:40:56.244 192.168.1.20 - 192.168.1.1 - ICMP 100 Bytes
312 07:40:55.842 192.168.1.20 7144 185.199.108.153 8080 TCP 745 Bytes
311 07:40:55.443 10.0.0.1 - 192.168.1.1 - ARP 364 Bytes

Module 2:Packet Analysis and Visualization

- This module receives the packets captured by the Packet Capture module.
- It extracts important information such as source IP, destination IP, protocol, packet size, and timestamp.
- It identifies and classifies protocols such as TCP, UDP, ICMP, and ARP.
- The packet information is processed and organized so that users can easily understand the network traffic.
- The analyzed information is displayed through the graphical user interface for easy monitoring.

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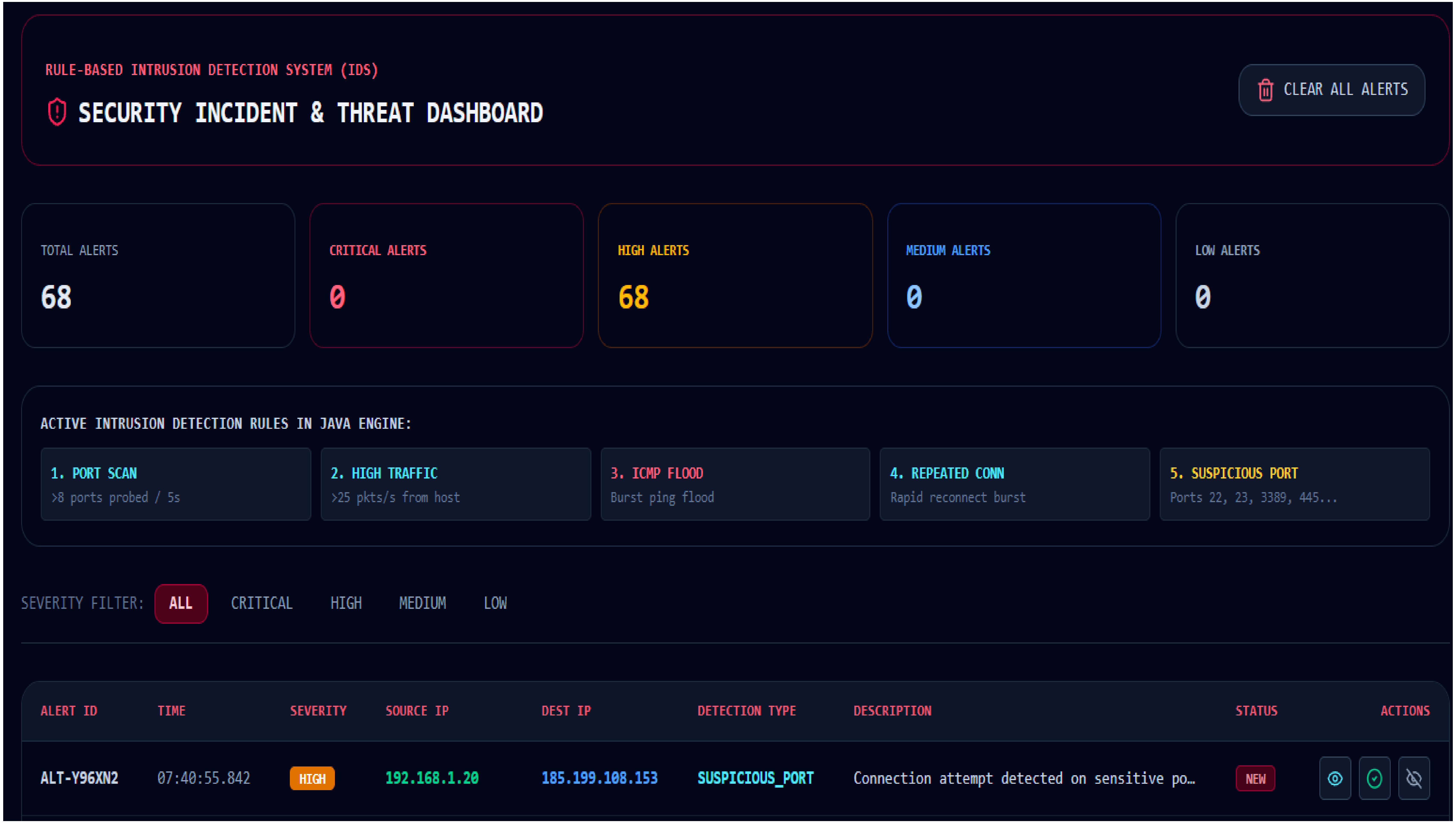
TRAFFIC ANALYTICS & DEEP PACKET INSPECTION
NETWORK PACKET VISUALIZATION DASHBOARD
TOTAL PACKETS
316
TCP PACKETS
138
UDP PACKETS
103
ICMP PACKETS
39
ARP PACKETS
36
AVG FRAME SIZE
525.2 B
TRAFFIC RATE
2.4 P/s
1. PROTOCOL DISTRIBUTION
2. TRAFFIC RATE OVER TIME
16
12
8
4
0
07:40:49
07:40:51
07:40:52
07:40:53
07:40:54
07:40:55
3. PACKET SIZE ANALYSIS
180
135
90
45
0
<64B
64-256B
256-512B
512-1024B
>1024B

Module 3: Network Monitoring Dashboard

- This module presents the analyzed network information in a simple graphical dashboard.
- It displays captured packet statistics and important network traffic information.
- It helps users observe different protocols and network activity.
- The dashboard makes it easier to understand the current network communication and traffic flow.
- It provides a user-friendly view for continuous network performance monitoring.


RULE-BASED INTRUSION DETECTION SYSTEM (IDS)
SECURITY INCIDENT & THREAT DASHBOARD
CLEAR ALL ALERTS
TOTAL ALERTS
68
CRITICAL ALERTS
0
HIGH ALERTS
68
MEDIUM ALERTS
0
LOW ALERTS
0
ACTIVE INTRUSION DETECTION RULES IN JAVA ENGINE:
1. PORT SCAN
>8 ports probed / 5s
2. HIGH TRAFFIC
>25 pkts/s from host
3. ICMP FLOOD
Burst ping flood
4. REPEATED CONN
Rapid reconnect burst
5. SUSPICIOUS PORT
Ports 22, 23, 3389, 445...
SEVERITY FILTER:
ALL
CRITICAL
HIGH
MEDIUM
LOW
ALERT ID
TIME
SEVERITY
SOURCE IP
DEST IP
DETECTION TYPE
DESCRIPTION
STATUS
ACTIONS
ALT-Y96XN2
07:40:55.842
HIGH
192.168.1.20
185.199.108.153
SUSPICIOUS_PORT
Connection attempt detected on sensitive po...
NEW

AUDIT REPORT GENERATION ENGINE

SESSION SUMMARY & COMPLIANCE REPORT

Export CSV | Export JSON | **Print Report**

NETWORK PACKET SNIFFER & IDS REPORT

Real-Time Packet Capture, Analysis & Intrusion Detection

Report Date: 8/21/2026, 1:43:09 PM
Generated by: Java Spring Boot Pcap4J Engine

1. CAPTURE SESSION DETAILS

Selected Interface:	Start Time:	End Time:	Total Packets:
eth0	**1:10:24 PM**	**Ongoing Session**	**316**

2. PROTOCOL BREAKDOWN STATISTICS

TCP Packets	UDP Packets	ICMP Packets	ARP Packets
138	103	39	36

TOP SOURCE HOST IPS

185.199.108.153	33 Packets (10%)
142.250.190.46	32 Packets (10%)
10.0.0.1	28 Packets (9%)
192.168.1.20	27 Packets (9%)
192.168.1.1	26 Packets (8%)

TOP DESTINATION IPS

192.168.1.1	66 Packets (21%)
142.250.190.46	35 Packets (11%)
10.0.0.42	30 Packets (9%)
192.168.1.10	27 Packets (9%)
10.0.0.1	26 Packets (8%)

Technologies Used

Software

- Java
- Java Swing / JavaFX – Graphical User Interface
- Npcap – Packet capture support
- Packet Capture Library – For capturing network packets
- IDE: IntelliJ IDEA / Eclipse / VS Code

Concepts

- Network Packet Capture
- Protocol Analysis
- TCP/IP Networking
- Real-Time Monitoring
- Data Visualization

Advantage

- Real-time packet monitoring.
- Easy identification of network protocols.
- Displays important packet information clearly.
- Helps in basic network troubleshooting.
- Provides a simple and user-friendly interface.
- Can be used for learning and understanding network traffic.

Future Scope

- Implement advanced packet filtering.
- Add packet logging and report generation.
- Add graphical charts for traffic analysis.
- Implement bandwidth and traffic-rate monitoring.
- Add automated anomaly detection.
- Integrate advanced intrusion detection features in future versions.

Conclusion

- The project demonstrates a Java-based approach to network performance monitoring.
- It captures and analyzes network packets in real time.
- Important packet information such as IP addresses, protocols, packet size, and timestamps is displayed.
- Protocol classification makes network traffic easier to understand.
- The graphical interface provides a simple way to monitor network communication.

Reference

1. R. Doriguzzi-Corin, L. A. D. Knob, L. Mendozzi, D. Siracusa, and M. Savi, "Introducing Packet-Level Analysis in Programmable Data Planes to Advance Network Intrusion Detection," *Computer Networks*, 2024.

2. 2. H. J. Jin, F. R. Ghashghaei, N. Elmrabit, Y. Ahmed, and M. A. Hossain, "Enhancing Sniffing Detection in IoT Home Wi-Fi Networks: An Ensemble Learning Approach with Network Monitoring System," *IEEE Access*, 2024.

3. J. Ghadermazi, A. Shah, and N. Bastian, "Towards Real-Time Network Intrusion Detection With Image-Based Sequential Packets Representation," *IEEE Transactions on Big Data*, 2024.

THANK YOU

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